Submission D solution to Problem 7

Problem Statement

Let $G = \langle a_1, \dots, a_r \rangle$ be a free group. A word on the letters $L = \{a_1, a_1^{-1}, \dots, a_r, a_r^{-1}\}$ is called reducible if it equals $1 \in G$. A word is called quasi-reduced if it does not contain three consecutive letters $\ell\ell^{-1}\ell$ for any $\ell \in L$.

Fix a reducible quasi-reduced word $w = w_1 \cdots w_{2n}$ (with $w_i \in L$). A matching M consists of n intervals, each properly embedded in $\mathbb{R} \times \mathbb{R}_+$, satisfying the following:

- Any two intervals intersect transversally, and there are no triple points.
- $\partial M = \{1, \dots, 2n\} \times 0 \subset \mathbb{R} \times 0 \subset \mathbb{R} \times \mathbb{R}_+$.
- For each component R of $\mathbb{R} \times \mathbb{R}_+ \setminus M$, and for each component A of $M \cap \partial \overline{R}$, ∂A consists of some two points $(i, 0)$ and $(j, 0)$ with $w_i = w_j^{-1}$.

A k-crossing matching is a matching where the total number of self-intersections is k.

Let F_w be the following CW complex whose k-cells correspond to isotopy classes of k-crossing matchings. Given a k-crossing matching M with crossing set $X(M)$, the associated k-cell is given by $C(M) \coloneqq \prod_{c \in X(M)} I_c$, where I_c is an interval whose endpoints correspond to the two ways of locally resolving M near c. The attaching maps are given as follows: On a facet of the form

$$a \times \prod_{d \in X(M) \setminus \{c\}} I_d, \qquad c \in X(M), a \in \partial I_c,$$

let M_a be the $(k-1)$-crossing matching obtained by resolving M at c according to a. Then the $(k-1)$-cell associated to M_a is given by $C(M_a) = \prod_{d \in X(M_a)} I_d = \prod_{d \in X(M) \setminus \{c\}} I_d$, and the above facet maps to this $(k-1)$-cell canonically.

Is F_w contractible?

Solution

Theorem 1. *For a reducible, quasi-reduced word w, the CW complex F_w is not necessarily contractible.*

Proof. **1. Counterexample Construction**

Let $G = \langle x, y, z, u \rangle$ be a free group. Define the 12-letter word:

$$w = x_1\, y_2\, y_3^{-1}\, x_4^{-1}\, x_5\, z_6\, z_7^{-1}\, x_8^{-1}\, x_9\, u_{10}\, u_{11}^{-1}\, x_{12}^{-1}$$

The word reduces to $1 \in G$ by canceling adjacent unique pairs (yy^{-1}), (zz^{-1}), (uu^{-1}) and sequentially collapsing the remaining xx^{-1} pairs. Because the strictly alternating x sequence is separated by uniquely labeled pairs, no continuous substring of the form $\ell\ell^{-1}\ell$ exists, ensuring w is strictly quasi-reduced.

2. Absolute Cap Isolation via Tree Perimeters

Let M be any valid k-crossing matching in F_w. First, M cannot contain a topological cycle. If M contained a closed loop, the innermost cycle would bound an internal planar region R_{int} disjoint from the real axis. This yields an intersection component $A = M \cap \partial\overline{R_{\text{int}}}$ that is a closed loop, explicitly possessing an empty boundary ($\partial A = \emptyset$). This strictly violates the CW complex constraint that ∂A must consist of exactly two boundary points. Therefore, M is necessarily a planar forest.

For any tree component $T \subset M$, the topological boundary of its regular complement neighborhood (the region boundaries of F_w) strictly traces an Euler perimeter tour that visits all leaves of T in a cyclic order. By the complex's condition, for any leaf v, the two region boundary components originating at v (representing the clockwise and counterclockwise directions of the perimeter tour) must terminate at a leaf paired with the inverse letter w_v^{-1}.

For the leaf at position 2, the letter is $w_2 = y$. The uniquely required inverse is at position 3 (y^{-1}). Therefore, the perimeter tour of the tree T containing 2 must visit 3 immediately in both the clockwise and counterclockwise directions. A tree whose cyclic leaf order dictates that 3 is adjacent to 2 on both sides must possess exactly two leaves, $\{2, 3\}$. Because internal arc crossings act strictly as degree-4 vertices (which inherently increase a connected component's leaf count to at least 4), a 2-leaf tree structurally cannot contain any crossings. Thus, the interval $(2, 3)$ is rigidly isolated and uncrossed in all valid matchings. Executing the exact same perimeter deduction independently for the unique inverse pairs at 6 and 10 structurally isolates the uncrossed intervals $(6,7)$ and $(10,11)$.

3. Enumeration of the Core Subcomplex

With the three padding caps isolated, the configuration space relies entirely on the 6 active core endpoints $\{1,4,5,8,9,12\}$. We relabel them $1' \ldots 6'$ carrying the alternating parities $x, x^{-1}, x, x^{-1}, x, x^{-1}$ (Odds are x, Evens are x^{-1}).

- **0-cells ($c_0 = 5$):** The valid 0-crossing matchings strictly correspond to the $C_3 = 5$ planar trees. Because the sequence parities alternately perfectly, all 5 validly pair x with x^{-1}.

- **1-cells ($c_1 = 6$):** A 1-cell contains exactly one crossing bridging two intervals (a,b) and (c,d) with endpoints $a < c < b < d$. The crossing defines four bounded regions whose perimeter traces connect the adjacent sequence of endpoints: $(a,d), (a,c), (c,b), (b,d)$. For these topological traces to correctly pair opposite parities, the 4 selected endpoints must strictly alternate in parity (O,E,O,E or E,O,E,O). Checking all $\binom{6}{4} = 15$ endpoint subsets reveals exactly 6 subsets alternate in parity. Resolving their crossings into planar states confirms that these 6 valid 1-cells completely connect the 5 non-crossing vertices to form exactly a complete bipartite $K_{2,3}$ 1-skeleton (two vertices of degree 3, three vertices of degree 2).

- **2-cells ($c_2 = 3$):** Exactly three valid subsets possess two crossings: $M_1 = \{(1',3'),(2',5'),(4',6')\}$, $M_2 = \{(1',5'),(2',4'),(3',6')\}$, and $M_3 = \{(1',4'),(2',6'),(3',5')\}$.

  To rigorously verify M_3 independent of geometric ambiguity, we execute an explicit Euler perimeter tour on its tree components. The intervals $\alpha = (1',4')$ and $\gamma = (2',6')$ cross at c_1; α and $\beta = (3',5')$ cross at c_2. Traversing the right-hand bounding walls of the tree's branches strictly turns at c_1 and c_2 to remain in the complement space. Originating from $1'$, the Euler tour visits the leaves in exactly the reverse ordered sequence: $1', 6', 5', 4', 3', 2'$. This cyclic boundary visitation proves the complement regions inherently pair $(1',6'), (6',5'), (5',4'), (4',3'), (3',2'), (2',1')$. Every region strictly pairs an Odd index with an Even index (opposite parities), completely satisfying the CW complex boundary conditions despite internal arc nestings. Executing the exact same tree Euler tour algorithm on M_1 and M_2 yields the identical cyclic leaf sequence. All three are valid 2-cells.

- **$\geq$ 3-cells ($c_{\geq 3} = 0$):** A 3-crossing matching on a 6-point space requires the 3 available intervals to mutually cross one another exactly once, strictly enclosing an internal central intersection triangle. As established in Step 2, any central region disjoint from the real axis inherently forces $\partial A = \emptyset$, which is strictly forbidden.

4. Topological Conclusion

The Euler characteristic of F_w is strictly calculated as:

$$\chi(F_w) = c_0 - c_1 + c_2 = 5 - 6 + 3 = 2$$

The Euler-Poincaré relation dictates the cellular homology via $\chi(F_w) = b_0 - b_1 + b_2$. Because the $K_{2,3}$ 1-skeleton is a connected graph, the 0-th Betti number is identically $b_0 = 1$. Substituting this yields $2 = 1 - b_1 + b_2$, rearranging to $b_2 = 1 + b_1$.

Because Betti numbers represent rank and are strictly non-negative ($b_1 \geq 0$), it structurally mandates $b_2 \geq 1$. Thus, the second homology group $H_2(F_w)$ is unconditionally non-trivial. A space is contractible only if all higher homology groups are strictly trivial; therefore, F_w is not contractible. $\square$